

22125096_HW08

Doan Cong Thanh

Set up

```
library(dplyr)
library(lmtest)
library(car)
setwd('D:/STAT/Lab/Lab8')
data<-read.csv('star.csv')
head(data)
```

```
##      id  schid    tchid tchexper absent  readscore  mathscore  totalscore  boy
## 1 10133 169280 16928003      7      5      427      478      905      1
## 2 10246 218562 21856202      8     28      450      494      944      0
## 3 10263 205492 20549204      3      2      483      513      996      0
## 4 10266 257899 25789904     12     10      456      513      969      1
## 5 10275 161176 16117602      2      3      411      468      879      1
## 6 10281 189382 18938204      7      2      443      473      916      1
##  white_asian black tchwhite tchmasters freelunch schurban schrural small
## 1          0      1          1          0          0          0          0      0
## 2          1      0          1          1          0          1          0      0
## 3          0      1          1          0          1          0          0      1
## 4          1      0          1          1          0          0          1      0
## 5          1      0          1          0          0          0          1      0
## 6          1      0          1          1          0          0          0      1
##  regular  aide
## 1          0      1
## 2          0      1
## 3          0      0
## 4          1      0
## 5          0      1
## 6          0      0
```

```
attach(data)
```

a)

```
regular_no_aide <- subset(data, regular == 1 & aide == 0)
regular_with_aide <- subset(data, regular == 1 & aide == 1)
smallClass <- subset(data, small == 1)
```

The average total score for (i) Students in regular-sized classrooms with full-time teachers, but no aide

```
avg_score_regular_no_aide <- mean(regular_no_aide$totalscore)
avg_score_regular_no_aide
```

```
## [1] 918.0429
```

(ii) students in regular-sized classrooms with full-time teachers, and an aide

```
avg_score_regular_with_aide <- mean(regular_with_aide$totalscore)
avg_score_regular_with_aide
```

```
## [1] NaN
```

(iii) students in small classrooms

```
avg_score_smallClass <- mean(smallClass$totalscore)
avg_score_smallClass
```

```
## [1] 931.9419
```

Observe about the test score

There is no student in regular class with an aide. Moreover, the small class has the average of total score higher than this one in regular class with no aide.

b)

```
model_b <- lm(totalscore ~ small + aide)
summary(model_b)
```

```
##
## Call:
## lm(formula = totalscore ~ small + aide)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -283.04  -50.94   -7.04   42.64  334.64
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  918.0429     1.6412  559.363 < 2e-16 ***
## small        13.8990     2.4085   5.771  8.3e-09 ***
## aide          0.3139     2.3102   0.136   0.892
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 73.49 on 5783 degrees of freedom
## Multiple R-squared:  0.0073, Adjusted R-squared:  0.006957
## F-statistic: 21.26 on 2 and 5783 DF,  p-value: 6.3e-10
```

Equation for estimated total score: $totalScore = 918.0429 + 13.899 \times small + 0.3139 \times aide$

This estimated total score is the same as the sample mean in question a (918.0429 for regular class with no aide, 931.9419 for small class).

$$\begin{cases} H_0 : \beta_3 = 0 \\ H_1 : \beta_3 \neq 0 \end{cases}$$

Apply Student's Test.

Assume $\alpha = 0.05$

Since n is quite large $\rightarrow$ Use p-value instead

Since p-value = 0.892 > 0.05

$\Rightarrow$ Fail to reject H_0

Hence, the aide variable is not useful.

c)

```
model_c <- lm(totalscore ~ small + aide + tchexper)
summary(model_c)
```

```
##
## Call:
## lm(formula = totalscore ~ small + aide + tchexper)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -274.13  -51.41   -7.44   42.57  341.53
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  904.7212     2.2280  406.071 < 2e-16 ***
## small        14.0061     2.3953   5.847 5.27e-09 ***
## aide        -0.6006     2.3065  -0.260  0.795
## tchexper      1.4690     0.1672   8.784 < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 73.08 on 5762 degrees of freedom
## (20 observations deleted due to missingness)
## Multiple R-squared:  0.02033,    Adjusted R-squared:  0.01982
## F-statistic: 39.86 on 3 and 5762 DF,  p-value: < 2.2e-16
```

$$\begin{cases} H_0 : \beta_4 = 0 \\ H_1 : \beta_4 \neq 0 \end{cases}$$

Apply Student's Test.

Assume $\alpha = 0.05$

Since n is quite large $\rightarrow$ Use p-value instead

Since p-value < 2×10^{-16} < 0.05

$\Rightarrow$ Reject H_0

Hence, tchexper variable is statistically significant.

We have:

in c) $\beta_2 = 14.0061, \beta_3 = -0.6006$

in d) $\beta_2 = 13.896, \beta_3 = 0.6983$

The addition of the TCHEXPER variable to the model does affect the estimates of β_2 and β_3 .

Specifically, it causes a slight increase in the estimated of β_2 and a substantial change in the estimated of β_3 , which changes from positive to negative

d)

```
model_d <- lm(totalscore ~ small + aide + tchexper + boy + freelunch + white_asian)
summary(model_d)
```

```
##
## Call:
## lm(formula = totalscore ~ small + aide + tchexper + boy + freelunch +
##     white_asian)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -255.27  -47.92   -8.07   40.29  323.48
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  923.2498    3.1210  295.822 < 2e-16 ***
## small        13.8960    2.2936   6.059 1.46e-09 ***
## aide          0.6983    2.2093   0.316  0.752
## tchexper      1.1140    0.1613   6.908 5.43e-12 ***
## boy        -14.0452    1.8457  -7.610 3.19e-14 ***
## freelunch   -34.1170    2.0639 -16.531 < 2e-16 ***
## white_asian  11.8373    2.2108   5.354 8.92e-08 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 69.98 on 5759 degrees of freedom
## (20 observations deleted due to missingness)
## Multiple R-squared:  0.1023, Adjusted R-squared:  0.1013
## F-statistic: 109.3 on 6 and 5759 DF,  p-value: < 2.2e-16
```

$$\begin{cases} H_0 : \beta_5 = 0 \\ H_1 : \beta_5 \neq 0 \end{cases}$$

Apply Student's Test.

Assume $\alpha = 0.05$

Since n is quite large $\rightarrow$ Use p-value instead

Since p-value = $3.19 \times 10^{-14} < 0.05$

$\Rightarrow$ Reject H_0

Hence, boy variable is statistically significant.

$$\begin{cases} H_0 : \beta_6 = 0 \\ H_1 : \beta_6 \neq 0 \end{cases}$$

Apply Student's Test.

Assume $\alpha = 0.05$

Since n is quite large $\rightarrow$ Use p-value instead

Since p-value $< 2 \times 10^{-16} < 0.05$

$\Rightarrow$ Reject H_0

Hence, freelunch variable is statistically significant.

$$\begin{cases} H_0 : \beta_7 = 0 \\ H_1 : \beta_7 \neq 0 \end{cases}$$

Apply Student's Test.

Assume $\alpha = 0.05$

Since n is quite large $\rightarrow$ Use p-value instead

Since p-value $= 8.92 \times 10^{-8} < 0.05$

$\Rightarrow$ Reject H_0

Hence, white_asian variable is statistically significant.

We have:

in c) $\beta_2 = 14.0061, \beta_3 = -0.6006$

in d) $\beta_2 = 13.896, \beta_3 = 0.6983$

These additions to the model do affect the estimates of β_2 and β_3 .

Specifically, it causes a slight decrease in the estimated of β_2 and a substantial change in the estimated of β_3 , which changes from negative to positive

e)

```
model_d <- lm(totalscore ~ small + aide + tchexper + boy + freelunch + white_asian + tchwhite + tchmast
summary(model_d)
```

```
##
## Call:
## lm(formula = totalscore ~ small + aide + tchexper + boy + freelunch +
##      white_asian + tchwhite + tchmasters + schurban + schrural)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -262.91  -47.12   -8.43   39.97  314.46
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  931.7553     3.9401  236.482 < 2e-16 ***
## small        13.9803     2.3023   6.072 1.34e-09 ***
## aide          1.0023     2.2166   0.452  0.65114
## tchexper      1.1562     0.1657   6.979 3.31e-12 ***
## boy         -14.0083     1.8426  -7.602 3.37e-14 ***
## freelunch   -32.5325     2.1260 -15.302 < 2e-16 ***
## white_asian  16.2326     2.7804   5.838 5.57e-09 ***
## tchwhite     -7.6683     2.8420  -2.698  0.00699 **
## tchmasters   -3.5598     2.0193  -1.763  0.07798 .
## schurban     -5.7499     2.8580  -2.012  0.04428 *
## schrural     -7.0061     2.5585  -2.738  0.00619 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 69.86 on 5755 degrees of freedom
```

```
## (20 observations deleted due to missingness)
## Multiple R-squared: 0.1059, Adjusted R-squared: 0.1043
## F-statistic: 68.14 on 10 and 5755 DF, p-value: < 2.2e-16
```

$$\begin{cases} H_0 : \beta_8 = 0 \\ H_1 : \beta_8 \neq 0 \end{cases}$$

Apply Student's Test.

Assume $\alpha = 0.05$

Since n is quite large $\rightarrow$ Use p-value instead

Since p-value = 0.00699 < 0.05 $\Rightarrow$ Reject H_0

Hence, tchwhite variable is statistically significant.

$$\begin{cases} H_0 : \beta_9 = 0 \\ H_1 : \beta_9 \neq 0 \end{cases}$$

Apply Student's Test.

Assume $\alpha = 0.05$

Since n is quite large $\rightarrow$ Use p-value instead

Since p-value = 0.07798 > 0.05 $\Rightarrow$ Fail to reject H_0

Hence, tchmasters variable is not useful

$$\begin{cases} H_0 : \beta_{10} = 0 \\ H_1 : \beta_{10} \neq 0 \end{cases}$$

Apply Student's Test.

Assume $\alpha = 0.05$

Since n is quite large $\rightarrow$ Use p-value instead

Since p-value = 0.04428 < 0.05 $\Rightarrow$ Reject H_0

Hence, schurban variable is statistically significant at significant level 95%.

$$\begin{cases} H_0 : \beta_{11} = 0 \\ H_1 : \beta_{11} \neq 0 \end{cases}$$

Apply Student's Test.

Assume $\alpha = 0.05$

Since n is quite large $\rightarrow$ Use p-value instead

Since p-value = 0.00619 < 0.05 $\Rightarrow$ Reject H_0

Hence, schrural variable is statistically significant at significant level 95%.

We have:

in d) $\beta_2 = 13.896, \beta_3 = 0.6983$

in d) $\beta_2 = 13.9803, \beta_3 = 1.0023$

These additions to the model do affect the estimates of β_2 and β_3 .

Specifically, it causes a slight increase in the estimated of β_2 and a slight increase in the estimated of β_3

f)

Part c) Analysis

In part (c), the variable tchexper (teacher experience) is added to the model:

Original model (part b): totalscore $\sim$ small + aide

$$\beta_2 = 13.899$$

$$\beta_3 = 0.3139$$

Model with tchexper (part c): totalscore $\sim$ small + aide + tchexper

$$\beta_2 = 14.0061$$

$$\beta_3 = -0.6006$$

Importance:

The change in β_3 dicates that the inclusion of teacher experience affects the interpretation of the aide's impact. The previously positive but insignificant effect of having an aide turns slightly negative, though still insignificant.

Teacher experience is statistically significant, suggesting that more experienced teachers contribute positively to the total score.

Part d) Analysis

In part (d), additional student-related variables (boy, freelunch, and white_asian) are added:

Model with tchexper and student-related variables: totalscore $\sim$ small + aide + tchexper + boy + freelunch + white_asian

$$\beta_2 = 13.896$$

$$\beta_3 = 0.6983$$

Importance:

Including student-related variables like boy, freelunch, and white_asian provides a more comprehensive model that accounts for student demographics and socioeconomic status, which significantly impact total scores.

These variables are statistically significant, showing they have a substantial effect on student performance. For instance, being a boy or receiving free lunch (indicative of lower socioeconomic status) negatively affects scores, while being white or Asian has a positive impact.

Part e) Analysis

In part (e), additional teacher and school-related variables (tchwhite, tchmasters, schurban, and schrural) are added:

Model with all variables: totalscore $\sim$ small + aide + tchexper + boy + freelunch + white_asian + tchwhite + tchmasters + schurban + schrural

$$\beta_2 = 13.9803$$

$$\beta_3 = 1.0023$$

Importance:

The inclusion of these variables indicates that teacher and school characteristics also play a role in influencing student scores.

The statistical significance of these variables highlights their importance in understanding the broader context of student performance. For instance, being in an urban or rural school, or having a white teacher, significantly affects scores.

Conclusion

Teacher experience in part (c) significantly impacts student performance and slightly modifies the estimated effect of small classes and aides.

Student demographics and socioeconomic status in part (d) provide a more accurate model, showing that these factors have substantial impacts on performance and slightly adjust the treatment effects.

Teacher and school characteristics in part (e) further enhance the model, emphasizing the importance of considering a comprehensive set of factors when evaluating educational interventions.

g)

```
# Create a factor variable for school to include fixed effects
data$school <- as.integer(schrural == 1 | schurban == 1)
str(data)
```

```
## 'data.frame': 5786 obs. of 20 variables:
## $ id : int 10133 10246 10263 10266 10275 10281 10282 10285 10286 10287 ...
## $ schid : int 169280 218562 205492 257899 161176 189382 189382 201449 230612 128068 ...
## $ tchid : int 16928003 21856202 20549204 25789904 16117602 18938204 18938203 20144901 23061203 ...
## $ tchexper : int 7 8 3 12 2 7 14 4 6 11 ...
## $ absent : int 5 28 2 10 3 2 7 8 2 17 ...
## $ readscore : int 427 450 483 456 411 443 448 463 472 428 ...
## $ mathscore : int 478 494 513 513 468 473 449 520 536 484 ...
## $ totalscore : int 905 944 996 969 879 916 897 983 1008 912 ...
## $ boy : int 1 0 0 1 1 1 1 0 1 1 ...
## $ white_asian: int 0 1 0 1 1 1 1 1 1 1 ...
## $ black : int 1 0 1 0 0 0 0 0 0 0 ...
## $ tchwhite : int 1 1 1 1 1 1 1 1 1 1 ...
## $ tchmasters : int 0 1 0 1 0 1 1 1 1 0 ...
## $ freelunch : int 0 0 1 0 0 0 1 1 0 0 ...
## $ schurban : int 0 1 0 0 0 0 0 0 0 0 ...
## $ schrural : int 0 0 0 1 1 0 0 1 1 1 ...
## $ small : int 0 0 1 0 0 1 0 1 0 0 ...
## $ regular : int 0 0 0 1 0 0 0 0 1 0 ...
## $ aide : int 1 1 0 0 1 0 1 0 0 1 ...
## $ school : int 0 1 0 1 1 0 0 1 1 1 ...
```

```
# Fit the base model (b) with school fixed effects
model_b <- lm(totalscore ~ small + aide + school, data = data)

# Fit the model (c) with TCHEXPER and school fixed effects
model_c <- lm(totalscore ~ small + aide + tchexper + school, data = data)

# Fit the model (d) with additional student variables and school fixed effects
model_d <- lm(totalscore ~ small + aide + tchexper + boy + freelunch + white_asian + school, data = data)

# Fit the model (e) with additional teacher and school variables and school fixed effects
model_e <- lm(totalscore ~ small + aide + tchexper + boy + freelunch + white_asian + tchwhite + tchmasters + school, data = data)

# Perform joint significance tests for the school fixed effects
# We use the `anova` function to compare models with and without school fixed effects

# Base model (b) without school fixed effects
model_b_no_school <- lm(totalscore ~ small + aide, data = data)

# Model (c) without school fixed effects
model_c_no_school <- lm(totalscore ~ small + aide + tchexper, data = data)

# Model (d) without school fixed effects
model_d_no_school <- lm(totalscore ~ small + aide + tchexper + boy + freelunch + white_asian, data = data)
```



```

# Model (e) without school fixed effects
model_e_no_school <- lm(totalscore ~ small + aide + tchexper + boy + freelunch + white_asian + tchwhite

# Coefficient of the model with school
# The base model (b) with school fixed effects
coef(model_b)

## (Intercept)      small      aide      school
## 931.92663542  13.33917367  -0.05459716 -17.33306601

# The base model (c) with school fixed effects
coef(model_c)

## (Intercept)      small      aide      tchexper      school
## 918.3121899  13.4740395  -0.9435396   1.4146434 -16.3517975

# The base model (d) with school fixed effects
coef(model_d)

## (Intercept)      small      aide      tchexper      boy      freelunch
## 928.1495234  13.6526702   0.5129686   1.0926840 -14.0294067 -32.4372621
## white_asian      school
## 12.5783199  -7.5101501

# The base model (e) with school fixed effects
coef(model_e)

## (Intercept)      small      aide      tchexper      boy      freelunch
## 931.755252  13.980269   1.002339   1.156161 -14.008255 -32.532461
## white_asian      tchwhite      tchmasters      schurban      schrural      school
## 16.232640  -7.668267  -3.559812  -5.749880  -7.006102      NA

# Compare models with and without school fixed effects
anova_b <- anova(model_b_no_school, model_b)
anova_c <- anova(model_c_no_school, model_c)
anova_d <- anova(model_d_no_school, model_d)
anova_e <- anova(model_e_no_school, model_e)

# Print the results
# Joint significance test for model (b)
anova_b

## Analysis of Variance Table
##
## Model 1: totalscore ~ small + aide
## Model 2: totalscore ~ small + aide + school
##   Res.Df      RSS Df Sum of Sq    F    Pr(>F)
## 1    5783 31232400
## 2    5782 30938125  1    294275 54.997 1.381e-13 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```

```
# Joint significance test for model (c)
anova_c
```

```
## Analysis of Variance Table
##
## Model 1: totalscore ~ small + aide + tchexper
## Model 2: totalscore ~ small + aide + tchexper + school
##   Res.Df      RSS Df Sum of Sq    F    Pr(>F)
## 1    5762 30777099
## 2    5761 30516027  1    261072 49.287 2.466e-12 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
# Joint significance test for model (d)
anova_d
```

```
## Analysis of Variance Table
##
## Model 1: totalscore ~ small + aide + tchexper + boy + freelunch + white_asian
## Model 2: totalscore ~ small + aide + tchexper + boy + freelunch + white_asian +
##   school
##   Res.Df      RSS Df Sum of Sq    F    Pr(>F)
## 1    5759 28203499
## 2    5758 28151676  1    51823 10.6 0.001138 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
# Joint significance test for model (e)
anova_e
```

```
## Analysis of Variance Table
##
## Model 1: totalscore ~ small + aide + tchexper + boy + freelunch + white_asian +
##   tchwhite + tchmasters + schurban + schrural
## Model 2: totalscore ~ small + aide + tchexper + boy + freelunch + white_asian +
##   tchwhite + tchmasters + schurban + schrural + school
##   Res.Df      RSS Df Sum of Sq F Pr(>F)
## 1    5755 28089838
## 2    5755 28089838  0          0
```

For all models (b, c, d, e), the F-test results show extremely significant p-values ($p < 0.05$), demonstrating that the inclusion of school fixed effects is statistically significant in every model.

Impact on Coefficients β_2 and β_3 :

β_2 : The coefficient for SMALL remains consistently significant and positive across all models, with only slight changes in magnitude after adding fixed effects. This indicates that the impact of being in a smaller class on the total score is strong and significant.

β_3 : The coefficient for AIDE remains insignificant in all models, suggesting that having an aide does not significantly influence the total score. The inclusion of school fixed effects does not notably change this estimate.